

Virang Parekh

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EDUCATION

Albert-Ludwigs-Universität-Freiburg | DE

Apr. 2026 – Present

Master of Science in Informatik

Current Courses: SAT Solvers, Probabilistic Graphical Models, Reinforcement Learning

D. J. Sanghvi College of Engineering, University of Mumbai | IN

Aug. 2018 – May 2022

Bachelors of Engineering in Computer Engineering

9.00 CGPA

Core Courses: Data Structures and Algorithms, Database Management Systems, Operating Systems, Cryptography & System Security, Digital Signals & Image Processing, Machine Learning

EXPERIENCE

R&D Engineer

June 2022 – Present

Samsung Research

Noida, IN

Android Application

June 2022 – July 2025

- Developing and maintaining service framework modules for the Samsung Health App.
- Contributed to the development of End-to-End Encryption (E2EE) for the Samsung Galaxy S24.
- Developed support for E2EE in devices targeting Android 13 and lower.
- Worked with AbstractThreadedSyncAdapter, WorkManager, Cloud Messaging and Coroutines
- Integrated SDKs from stakeholders, including SCloud, Knox Matrix and Samsung Account Client

iOS Development

July 2025 – February 2026

- Built the Samsung Health iOS Data Service Framework using asynchronous, recoverable APIs.
- Engineered a reliable DAPI pipeline for seamless health data reconciliation with SCloud.
- Developed a native HealthData SDK to enable secure third-party metric integrations.
- Designed a custom push notification delegation system for precise, scalable alert delivery.

Data Analyst Intern

June 2021 – November 2021

Tech Mahindra

Remote

- Worked on a Multi-Modal Transportation based project for a 'Pune' specific use case
- Worked on Data Wrangling, Warehousing, Data Modeling, DAX and Querying
- Built simple, lightweight, interactive and easy-to-understand dashboards for Visualizations.

Software Development and Data Analytics Intern

September 2020 – December 2020

Senquire

Pune, IN

- Created a machine vision system for real-time machinery part condition monitoring.
- Implemented multiprocessing, data preprocessing, queuing, and designed a lightweight interface.

RESEARCH AND PUBLICATIONS

Performance Analysis of Various Asymmetric Public-key Cryptosystem | ICPCSN 2022

[Springer Link](#)

Video-based Micro Expressions Recogn. using Deep and Transfer Learning | ICDSAI 2022

[Springer Link](#)

PROJECTS

Incisive Analysis Of EQ Using AV Data | Deep Learning, Transfer Learning and Bi-LSTMs

Major Project

- Investigated how the MSCEIT and the STAR method work for the measurement of EQ.
- Proposed a succinct approach to automate and digitize the manual process for the same

OneStop Fashion | Django, TensorFlow, OpenCV-2

[GitHub Link](#)

- A Reverse Image detection system which from the image detects what type of fashion apparel it is.
- Worked on designing the database, web scraping bots and integrating ML models with the software.

StudVisor | Django REST Framework, Sci-kit Learn

[GitHub Link](#)

- A web based recommendation system which predicts a student's placement opportunity.
- Designed the database, built REST APIs and ML models for prediction and recommendation.

POSITIONS OF RESPONSIBILITIES

Placement Coordinator

August 2021 – May 2022

Dwarkadas J. Sanghvi College of Engg.

Mumbai, IN

- Assisted Computer Engineering students with job search logistics and support.
- Coordinated between students and recruiters to facilitate a smooth interview process.
- Managed the placement database with student profiles, job stats as well as recruiter feedback.

Unicode

September 2019 – May 2022

Dwarkadas J. Sanghvi College of Engg.

Mumbai, IN

- Worked on a solution for managers and salespeople to efficiently track deliverables and stock levels.
- Developed backend REST APIs for location tracking and bill management.
- Organized a seminar for Junior Year students for graduate studies and job prospects

EXTRA CIRRICULARS

Library Secretary

July 2019 – June 2020

Shri Mahavir Jain Vidyalyaya

Mumbai, IN

- Edited and moderated content of the institute's annual magazine
- Worked as a library secretary, promoting Fine and Performing Arts
- Started a weekly book lover's club, trying to inculcate the habit of reading among students

DJS Aura

September 2018 – December 2020

Dwarkadas J. Sanghvi College of Engg.

Mumbai, IN

- Member of the college's official drama club
- Wrote and performed Street plays around social issues like child abuse and Domestic Violence against women
- Participated and won multiple street play competitions in and around Mumbai.

CERTIFICATIONS

Programming Languages & Fundamentals

- Software Competency (SWC) Test - Professional Grade
- Software Competency (SWC) Test - Advance Grade
- Learn Java Cryptography

Software Development & Architecture

- Software Design: Modelling with UML
- Complete Kotlin Design Patterns
- Advanced Design Patterns
- Using SOLID Principles to Write Better Code

Machine Learning

- Applied Machine Learning - Foundations, Feature Engg., Ensemble Learning
- Computer Vision for Data Scientists
- Artificial Intelligence - Intermediate Level Training

Non-Technical & Leadership

- Jeff Dyer on Innovation
- Leadership Foundations: Leadership Styles and Models
- Understanding Sustainability Management
- Communicate with Emotional Intelligence
- Managing Anxiety in the Workplace

SKILLS & TECHNOLOGIES

Programming Languages: C, Java, Kotlin, Python and JavaScript

Source Control & DevOps: Git, Perforce Helix, Docker, Heroku, Gradle

Android Development: MVVM, Room DB, Coroutines, WorkManager, Dagger (DI), SDK Dev, SyncAdapter, Firebase

Web Frontend Development: HTML, CSS, Bootstrap, React JS (Beginner)

Web Backend Development: Django, RESTful API Development, Cookie-cutter Django, Celery

Artificial Intelligence: TensorFlow, Pandas, Numpy, scikit-learn, OpenCV-2

Others: SQLite DB, PostgreSQL, Power BI, Plotly, UiPath (VB.NET), Scrapy